

Adaptive Leverage v2.0

From score-counting to state-aware leverage

Core idea: V2 keeps the same three market signals as V1, but stops treating every two-signal combination as equivalent. It preserves signal identity and adds one simple trend-slope gate to size the strongest recovery regime.

What stayed the same

- Same three inputs: 21/200-day moving-average ratio, 65/252-day volatility ratio, and price-to-prior-252-day-high ratio.
- Same signal thresholds: SMA ratio > 1.00, vol ratio < 1.00, and price/high ratio > 0.90.
- Same one-day trading lag: all information used for today's position is lagged one trading day.
- Same leverage ceiling: maximum target exposure remains 3.0x equity.

The architectural change

V1	V2
Compresses signals into a count. Score = SMA + VOL + P2H.	Preserves the exact signal state. State is written as SMA / VOL / P2H, e.g. 110.
Every score-2 day receives 0.9x, whether the active pair is 011, 101, or 110.	011, 101, and 110 can receive different exposure because their historical behavior was materially different.
Simple and robust, but discards information about which signals are on.	Uses the same information more efficiently without adding an unrelated fourth indicator.

V2 exposure map

State	Interpretation	V1 exposure	V2 exposure
000	No signals on	0.0x	0.0x
001	P2H only	0.5x	0.5x
010	Low-vol only	0.5x	0.5x
100	Trend only	0.5x	0.0x
011	Low-vol + near high	0.9x	0.5x
101	Trend + near high	0.9x	0.9x
110	Trend + low-vol, below 90% of prior high	0.9x	1.5x or 3.0x*
111	All three signals on	3.0x	3.0x

*For state 110, V2 uses 3.0x when the 200-day moving average is above its level 20 trading days earlier; otherwise it uses 1.5x.

Why V2 was introduced

The original score worked, but research showed that the identity of the active signals mattered more than the raw count. In particular, the three score-2 states did not behave alike. The 110 state - positive trend and relatively low volatility while still more than 10% below the prior high - repeatedly behaved like a strong recovery regime, while 011 was much weaker.

Research observation	Design response in V2
Score 2 mixed together 011, 101, and 110.	Preserve the full three-bit state instead of only the score.
110 historically supported more exposure than the other score-2 states.	Raise 110 from 0.9x to at least 1.5x.
Some 110 episodes were healthy recoveries; others were false starts.	Require a rising long-term trend before moving 110 all the way to 3.0x.

The new 110 slope gate

Unconfirmed recovery: $110 + \text{non-rising } 200DMA = 1.5x$. The state is favorable enough to exceed V1's 0.9x, but it does not receive maximum leverage.

Confirmed recovery: $110 + \text{rising } 200DMA = 3.0x$. The long-term trend is now improving, so V2 treats the regime more like the fully bullish 111 state.

Implementation changes

- Tradeable weights are explicit. Exposure below 1.0x is cash + SPY; exposure from 1.0x to 3.0x is implemented with SPY + UPRO. Example: $1.5x = 75\% \text{ SPY} + 25\% \text{ UPRO}$.
- Cash is no longer assumed to earn zero. The tradeable backtest uses a lagged 13-week T-bill yield proxy ($\wedge IRX$).
- Transaction costs are configurable. The script can charge basis points per dollar traded.
- V1 remains intact. The script reports and exports V1 and V2 separately so V1 remains the permanent control.

Research backtest snapshot

Historical sample	V1	V2 slope-gated	Change
Synthetic, 2000-2026	14.35% CAGR 0.625 Sharpe -45.1% max DD	15.58% CAGR 0.659 Sharpe -45.8% max DD	+1.23% CAGR Higher Sharpe Similar DD
Actual SPY/UPRO/T-bill, 2010+	24.05% CAGR 0.874 Sharpe -37.2% max DD	24.98% CAGR 0.901 Sharpe -37.6% max DD	+0.93% CAGR Higher Sharpe Similar DD
Same ETF test, 10 bps trading cost	22.55% CAGR 0.833 Sharpe	23.43% CAGR 0.858 Sharpe	+0.88% CAGR Higher Sharpe

Historical results are research backtests, not forecasts. Results vary by start date, data source, ETF availability, transaction-cost assumptions, taxes, and implementation details.

Bottom line

V2 is not a new signal strategy. It is a more information-efficient version of V1. The inputs and core thresholds remain the same; the main change is that V2 recognizes that different signal combinations represent different market regimes and sizes them accordingly. The 200DMA slope gate is used only to decide whether the 110 recovery state deserves moderate or maximum leverage.

Main caveat: V2 is still a leveraged equity strategy. Its incremental return has historically been concentrated in a limited number of strong recovery episodes. The slope gate reduces false-recovery exposure, but it is not a crash-protection mechanism.